

Standard form in action

Sometimes it is difficult to compare extremely large or small numbers because of the number of digits they contain. Standard form makes this easier.

The mass of Earth is 5,974,200,000,000,000,000,000 kg

24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1
 5,974,200,000,000,000,000,000.0 kg

The decimal point moves **24 places** to the left.

The mass of the planet Mars is

23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1
 641,910,000,000,000,000,000.0 kg

The decimal point moves **23 places** to the left.

Written in standard form these numbers are much easier to compare. Earth's mass in standard form is

$$5.9742 \times 10^{24} \text{ kg}$$

The mass of Mars in standard form is

$$6.4191 \times 10^{23} \text{ kg}$$



► Comparing planet mass

It is immediately evident that the mass of the Earth is bigger than the mass of Mars because 10^{24} is 10 times larger than 10^{23} .

EXAMPLES OF STANDARD FORM

Example	Decimal form	Standard form
Weight of the Moon	73,600,000,000,000,000,000 kg	$7.36 \times 10^{22} \text{ kg}$
Humans on Earth	6,800,000,000	6.8×10^9
Speed of light	300,000,000 m/sec	$3 \times 10^8 \text{ m/sec}$
Distance of the Moon from the Earth	384,000 km	$3.8 \times 10^5 \text{ km}$
Weight of the Empire State building	365,000 tons	$3.65 \times 10^5 \text{ tons}$
Distance around the Equator	40,075 km	$4 \times 10^4 \text{ km}$
Height of Mount Everest	8,850 m	$8.850 \times 10^3 \text{ m}$
Speed of a bullet	710 m/sec	$7.1 \times 10^2 \text{ m/sec}$
Speed of a snail	0.001 m/sec	$1 \times 10^{-3} \text{ m/sec}$
Width of a red blood cell	0.00067 cm	$6.7 \times 10^{-4} \text{ cm}$
Length of a virus	0.000 000 009 cm	$9 \times 10^{-9} \text{ cm}$
Weight of a dust particle	0.000 000 000 753 kg	$7.53 \times 10^{-10} \text{ kg}$

LOOKING CLOSER

Standard form and calculators

The exponent button on a calculator allows a number to be raised to any power. Calculators give very large answers in standard form.



△ **Exponent button**
 This calculator button allows any number to be raised to any power.

Using the exponent button:

4×10^2 is entered by pressing

4 × 10 X^y 2

On some calculators, answers appear in standard form:

$$1234567 \times 89101112 = 1.100012925 \times 10^{14}$$

so the answer is approximately
 110,001,292,500,000